Software Requirements Specification (SRS)

Project: Medi Core — Hospital Management System Version: 1.0
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Revision History

Authors	Date	Description	Version
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1. Introduction

1.1 Purpose

This SRS document specifies the functional and non-functional requirements for MediCore, a comprehensive Hospital Management System (HMS) designed to digitize hospital operations — from patient registration and appointment scheduling to billing, inventory.

1.2 Scope

MediCore is a hospital management System that enables patient management
, doctor scheduling, Department scheduling ,EMR handling, billing.

1.3 Definitions, Acronyms, and Abbreviations

HMS - Hospital Management System
EMR - Electronic Medical Record

1.4 References

SRS template (used as structural reference).

2. Overall Description

The system provides modules for patients, doctors, billing..

2.1 Product Perspective

The Hospital Management System (HMS) is a System that centralizes all hospital operations and data management. It stores patient data and provides
rolebased access for doctors, nurses, receptionists.

HMS is designed as a modular system — each functional module (e.g., Patient Management, Appointment Scheduling, Billing) can operate independently.

2.2 Product Functions

- Patient registration and management
- Appointment scheduling
- Doctor and Department management
- Billing modules

2.3 User Characteristics

The Hospital Management System is intended for use by various categories of hospital personnel, each with different technical backgrounds and responsibilities:

- Doctors: Use the system to view patient history, record diagnoses, prescribe medications, and request lab tests.
- Nurses: Record patient vitals, manage ward information, and update treatment progress.

Receptionists: Handle patient registration, appointment scheduling. process payments. Require experience with financial modules.

2.4 Operating Environment

Hardware:

Standard PCs, laptops, phones, tablets used within hospital departments

Software:

Operating System: Windows , Android, iOS, Linux

Network:

Local area network (LAN) within hospital premises..

2.5 Design and Implementation Constraints

- Must comply with healthcare data privacy regulations (e.g.,
- Must support multi-user concurrency and maintain data integrity under simultaneous operations.

- Limited by existing hospital IT infrastructure and hardware capabilities.

2.6 Assumptions and Dependencies

- Reliable network connectivity will be available for client-server communication.
- The hospital's policies for patient data confidentiality will align with the system's security model.
- The HMS depends on accurate input from staff; data quality directly affects system reliability.

3. Functional Requirements

- User Authentication and Roles
- Patient Management
- Appointment Scheduling
- Doctor and Staff Management
- Billing and Payment Management

FR-001 — User Authentication and Roles

- FR-001.1: The system shall allow users (admin, doctors, staff) to register and log in using a username and password.
- FR-001.2: The system shall authenticate users based on their role (e.g., Admin, Doctor, staff, Receptionist).
- FR-001.3: The system shall allow password recovery and account management.
- FR-001.4: The system shall restrict access to features based on user roles and permissions.

FR-002 — Patient Management

- FR-002.1: The system shall allow staff to add, edit, and delete patient records.
- FR-002.2: The system shall store patient personal details, medical history, diagnoses, and prescriptions.
- FR-002.3: The system shall allow searching for patients by name, ID, or contact information.
- FR-002.4: The system shall ensure that only authorized users can view or modify patient data.

FR-003 — Appointment Scheduling

- FR-003.1: The system shall allow Receptionist to schedule appointments with doctors.
- FR-003.2: The system shall display doctors’ available time slots.
- FR-003.3: The system shall send notifications/reminders for upcoming appointments.
- FR-003.4: The system shall allow rescheduling and cancellation of appointments.

- **FR-004 — Doctor and Staff Management**
- FR-004.1: The system shall allow admins to add, edit, or remove doctor and staff profiles.
- FR-004.2: Each profile shall include personal details, specialization, role, and contact information.
- FR-004.3: The system shall allow viewing assigned duties and schedules.

- **FR-005 — Billing and Payment Management**
- FR-005.1: The system shall allow staff to generate invoices for patient visits, tests, and procedures.
- FR-005.2: The system shall maintain records of all billing transactions.

Non-Functional Requirements

- Performance: Handle 200+ concurrent users.
- Security: hashed passwords.
- Usability: Simple UI for staff.
- Reliability: 99% uptime.

- **NFR-001 — Performance**
- The system shall handle 200+ concurrent users without performance degradation.

- **NFR-002 — Security**
- The system shall store hashed passwords and use secure communication.
- Only authorized users shall access sensitive data.

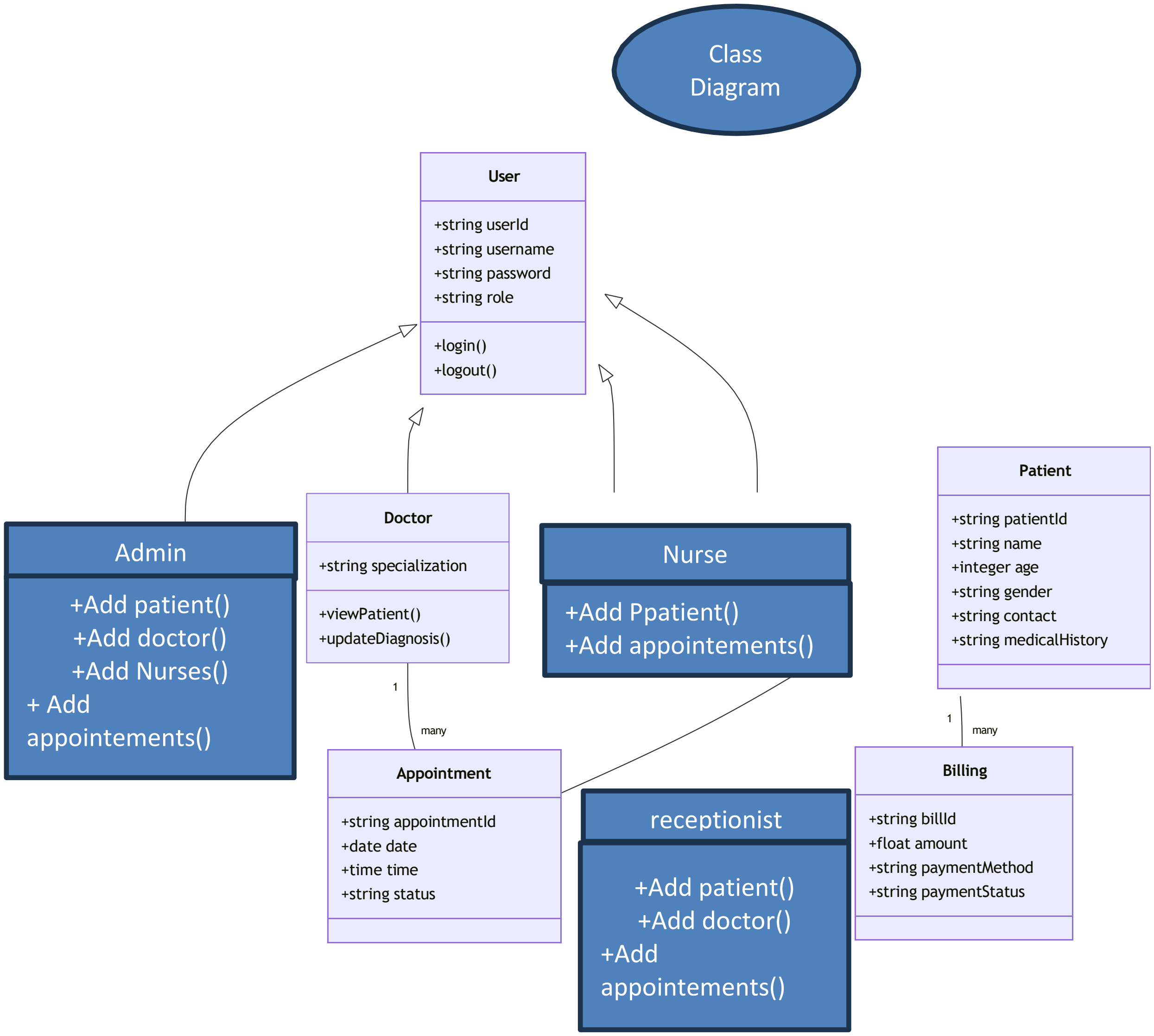
- **NFR-003 — Usability**
- The system should provide a simple and intuitive UI for doctors and staff.
- The interface should support desktops.

- **NFR-004 — Reliability**

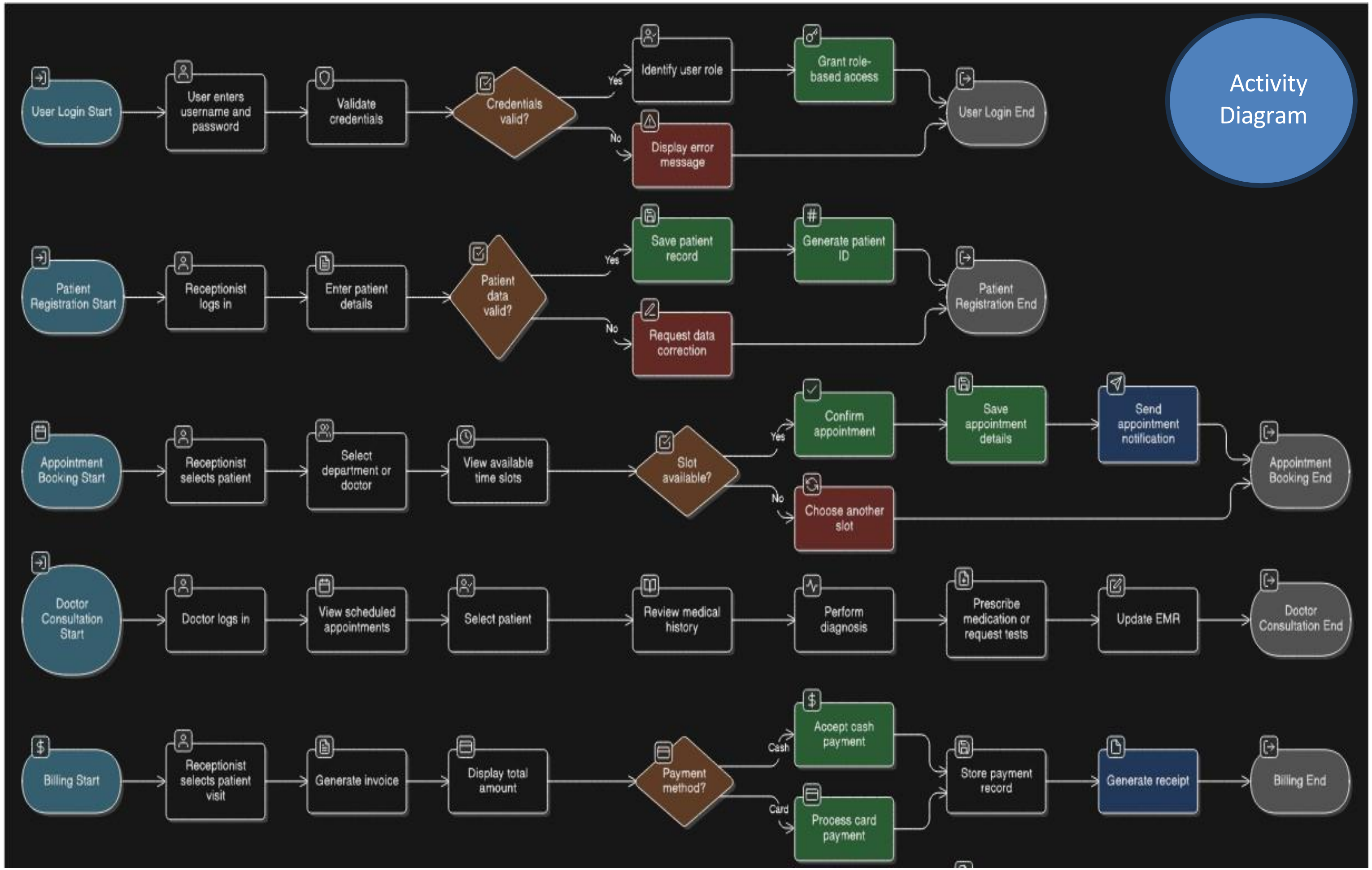
- The system shall maintain 99% uptime.

4. Use Cases

- Receptionist registers a new patient and schedules an appointment.
- Doctor views patient history and updates diagnosis.
- Receptionist generates a bill and marks payment received.

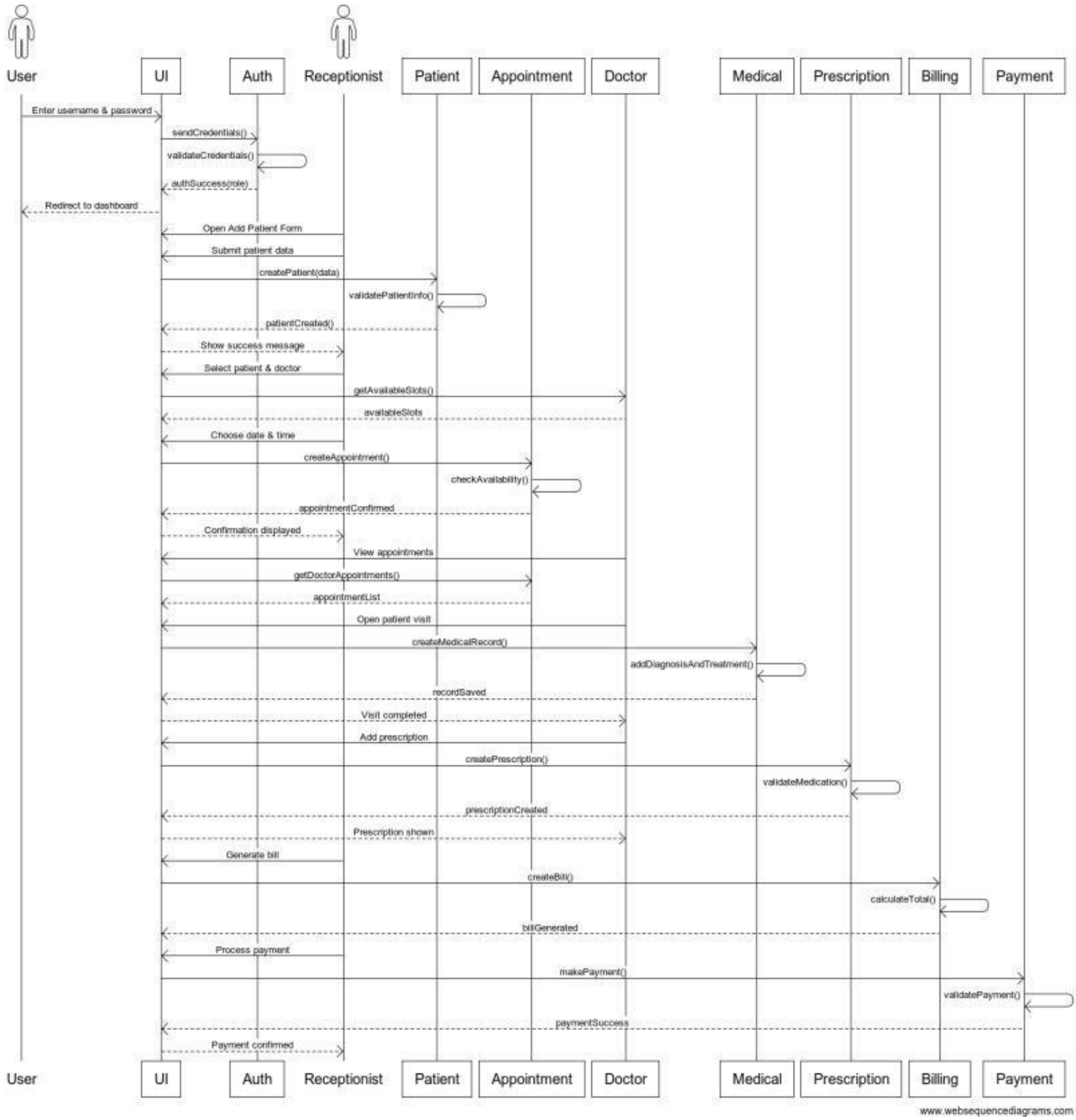


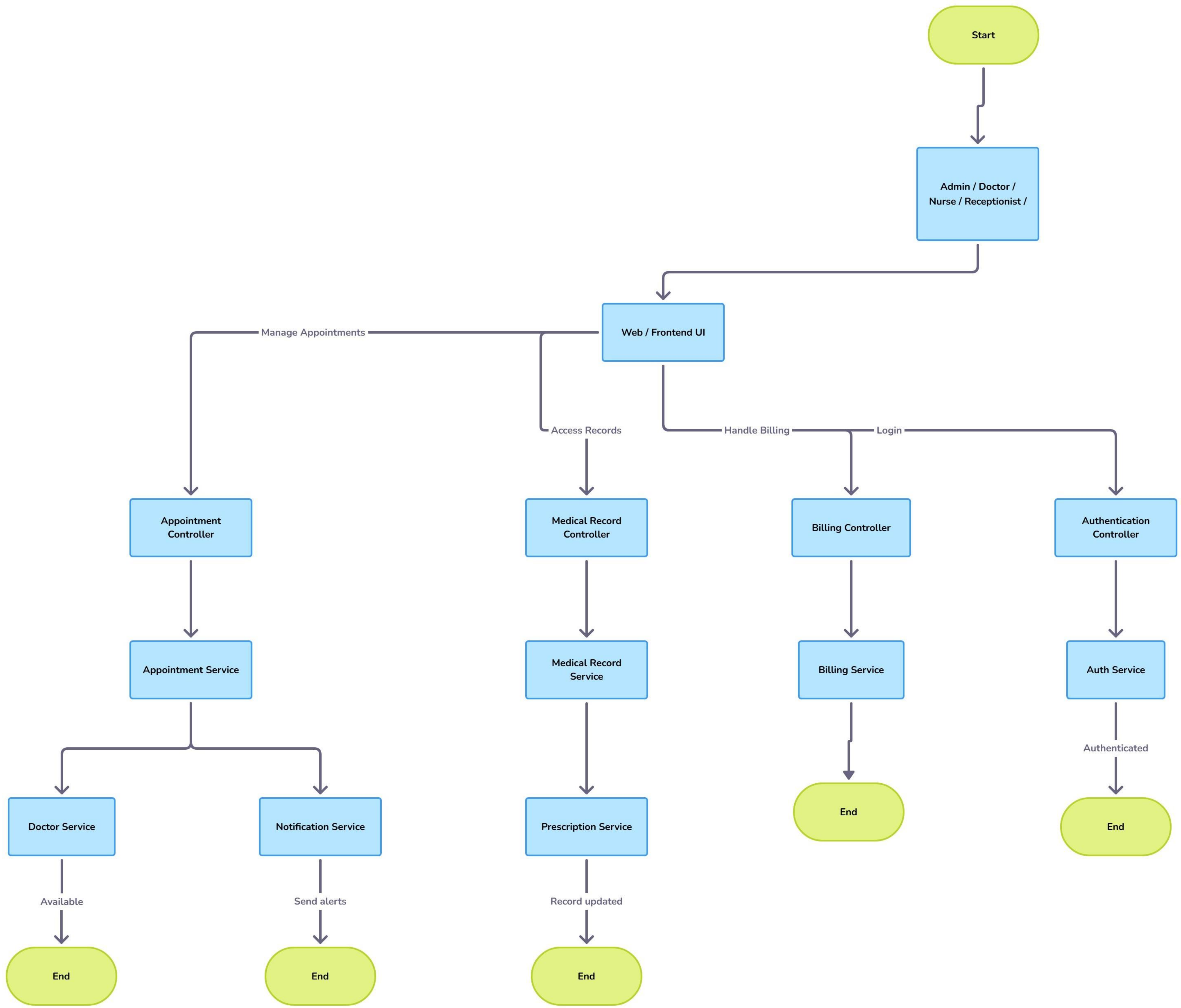
Class Diagram
User
+string userId
+string username
+string password
+string role
+login()
+logout()
Admin
+Add patient()
+Add doctor()
+Add Nurses()
+ Add appointements()
Doctor
+string specialization
+viewPatient()
+updateDiagnosis()
Nurse
+Add Ppatient()
+Add appointements()
Patient
+string patientId
+string name
+integer age
+string gender
+string contact
+string medicalHistory
1
many
Appointment
+string appointmentId
+date date
+time time
+string status
receptionist
+Add patient()
+Add doctor()
+Add appointements()
1
many
Billing
+string billId
+float amount
+string paymentMethod
+string paymentStatus



Activity Diagram
User Login Start
User enters username and password
Validate credentials
Credentials valid?
Yes
No
Identify user role
Grant role-based access
Display error message
User Login End
Patient Registration Start
Receptionist logs in
Enter patient details
Patient data valid?
Yes
No
Save patient record
Generate patient ID
Request data correction
Patient Registration End
Appointment Booking Start
Receptionist selects patient
Select department or doctor
View available time slots
Slot available?
Yes
No
Confirm appointment
Save appointment details
Send appointment notification
Choose another slot
Appointment Booking End
Doctor Consultation Start
Doctor logs in
View scheduled appointments
Select patient
Review medical history
Perform diagnosis
Prescribe medication or request tests
Update EMR
Doctor Consultation End
Billing Start
Receptionist selects patient visit
Generate invoice
Display total amount
Payment method?
Cash
Card
Accept cash payment
Process card payment
Store payment record
Generate receipt
Billing End

Sequence Diagram





Start
Admin / Doctor / Nurse / Receptionist /
Web / Frontend UI
Manage Appointments
Access Records
Handle Billing
Login
Appointment Controller
Medical Record Controller
Billing Controller
Authentication Controller
Appointment Service
Medical Record Service
Billing Service
Auth Service
End
Authenticated
End
Doctor Service
Notification Service
Prescription Service
Available
Send alerts
Record updated
End
End
End